CASC–NTP

Mathematical Architecture of the Geometric Cascade Machine

Configuration A: F–B–M–M–V–M–S

CASC Project

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Abstract

CASC–NTP is an autoregressive neural architecture with a recurrent causal carrier, a communication attention mechanism, a separate control attention mechanism, and a sequence of learned low-rank geometric operators. Each operator stage acts in a lifted operational space but returns its update to the hidden state before the next stage is evaluated. The next stage therefore conditions on the representation already transformed by previous stages. This document specifies the *machine itself*: its parameterization, tensor dimensions, recurrence, routing simplices, operator families, composition law, and lexical readout. A final section records the optimization mechanism used in the documented implementation: standard reverse-mode automatic differentiation (backpropagation) of next-token cross-entropy, followed by AdamW.

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1 Scope and mathematical conventions

The specified machine is **CASC–NTP, Configuration A**. It has four structurally identical blocks with independent trainable parameters. Each block processes a causal sequence, communicates recurrent memory into the hidden state, then applies seven consecutive geometric updates of families

$$(f_1, \dots, f_7) = (F, B, M, M, V, M, S). \tag{1}$$

The order is architectural: the three occurrences of M have their own operator parameters and cannot be treated as one shared metric operator.

Throughout, activations are *row vectors*; learned matrices multiply on the right. The position index is $i \in \{1, \dots, L\}$, block index is $b \in \{1, \dots, S\}$, geometric stage is $k \in \{1, \dots, K\}$, head index is $a \in \{1, \dots, H\}$, and operator slot is $j \in \{0, \dots, 5\}$. Unless otherwise specified, all formulas for a block omit the superscript (b) to keep the notation readable. Operators and parameters in distinct blocks are independent.

1.1 Dimensions

Symbol	Meaning	Configuration A
V	Vocabulary size	64,000
d_x	Hidden and lexical embedding width	1,024
d_g	Recurrent carrier / attention width	512
H	Number of heads in each attention	8
d_h	Width of each attention head ($Hd_h = d_g$)	64
D	Lifted operational dimension	4,096
r	Local operator rank	32
K	Geometric stages per block	7
S	Independent blocks	4
L	Causal sequence length	Configurable

Sequence length L is *not* fixed by the architecture. The training code defaults to $L = 512$. The published run used $L = 96$ for its initial training phase and an extension to $L = 384$ near the end. These are protocol choices, not different definitions of the machine.

1.2 Elementary functions and normalizations

For scalar or elementwise arguments, define

$$\sigma(t) = \frac{1}{1 + e^{-t}}, \qquad \operatorname{softplus}(t) = \log(1 + e^t), \qquad \operatorname{SiLU}(t) = t\sigma(t). \tag{2}$$

Hadamard multiplication is written $\odot$. The numerical stabilization constant is $\varepsilon = 10^{-6}$. For $x \in \mathbb{R}^d$, the parameter-free root-mean-square normalization is

$$\operatorname{rms}(x) = \frac{x}{\sqrt{\|x\|_2^2/d + \varepsilon}}. \tag{3}$$

The trainable version uses a learned coordinatewise gain $\gamma \in \mathbb{R}^d$:

$$\operatorname{RMSNorm}_d(x) = \gamma \odot \operatorname{rms}(x). \tag{4}$$

Different uses of RMSNorm have their own scale parameters unless parameter sharing is explicitly specified. We use

$$\Delta^{n-1} = \Big\{\alpha \in \mathbb{R}^n_{\geq 0} : \sum_{j=1}^{n} \alpha_j = 1\Big\} \tag{5}$$

for the $(n-1)$-dimensional probability simplex.

2 Lexical interface and block composition

Let $y_1, \dots, y_L \in \{0, \dots, V-1\}$ be token indices. The trainable embedding is

$$E \in \mathbb{R}^{V \times d_x}, \qquad E_{q\ell} \sim \mathcal{N}(0, 0.02^2), \qquad x_i^{[0]} = E[y_i] \in \mathbb{R}^{d_x}. \tag{6}$$

The same matrix is used by the final language-model head (weight tying).

With B_b denoting the complete transformation performed by block b, the machine body is

$$x^{[S]} = B_S \circ B_{S-1} \circ \cdots \circ B_1\big(x^{[0]}\big), \qquad S = 4. \tag{7}$$

Every block is internally divided into a *recurrent memory construction*, a *causal communication read*, and a *seven-stage controlled geometric cascade*. Shared parameters are shared *across the seven stages within a block*, not between distinct blocks.

3 Causal memory: signed-simplex GRU

3.1 Memory input and recurrent gates

At position i of a fixed block, write $x_i \in \mathbb{R}^{d_x}$ for the block input. Project it into the carrier width:

$$e_i = \mathrm{RMSNorm}_{d_x}(x_i) P_G \in \mathbb{R}^{d_g}, \quad P_G \in \mathbb{R}^{d_x \times d_g}, \quad \bar{g}_0 = 0. \tag{8}$$

P_G has no bias. The reset gate differs from the usual sigmoid GRU reset: it uses softplus and may therefore exceed one. With $W_r, W_z, W_h, U_r, U_z, U_h \in \mathbb{R}^{d_g \times d_g}$ and $b_r, b_z, b_h \in \mathbb{R}^{d_g}$,

$$r_i = \mathrm{softplus}(e_i W_r + \bar{g}_{i-1} U_r + b_r), \qquad r_i \in (0, \infty)^{d_g}, \tag{9}$$

$$z_i = \sigma(e_i W_z + \bar{g}_{i-1} U_z + b_z), \qquad z_i \in (0,1)^{d_g}, \tag{10}$$

$$h_i^{\mathrm{GRU}} = \tanh\big(e_i W_h + (r_i \odot \bar{g}_{i-1}) U_h + b_h\big), \tag{11}$$

$$g_i^{\mathrm{raw}} = (\mathbf{1} - z_i) \odot \bar{g}_{i-1} + z_i \odot h_i^{\mathrm{GRU}}. \tag{12}$$

The symbol h_i^{GRU} is the recurrent candidate, not the geometric-stage hidden state introduced below.

3.2 Signed-simplex factorization

The raw recurrence is converted into an oriented, normalized memory representation. Let β_{mem} and κ_{mem} denote the learnable scalars named β and κ in the code. Set, coordinatewise,

$$s_i = \tanh(\beta_{\mathrm{mem}} g_i^{\mathrm{raw}}), \qquad s_i \in (-1,1)^{d_g}, \tag{13}$$

$$m_{ij} = \sqrt{(g_{ij}^{\mathrm{raw}})^2 + \varepsilon}, \tag{14}$$

$$q_{ij} = \mathrm{softplus}(\kappa_{\mathrm{mem}} m_{ij}) > 0, \tag{15}$$

$$\alpha_{ij} = \frac{q_{ij}}{\sum_{\ell=1}^{d_g} q_{i\ell}}, \qquad \alpha_i \in \Delta^{d_g - 1}, \tag{16}$$

$$g_i = s_i \odot \alpha_i. \tag{17}$$

Thus the magnitude allocation α_i is a simplex, whereas g_i is a *signed simplex-weighted vector*; g_i is not itself a probability vector. Since $|s_{ij}| < 1$,

$$\|g_i\|_1 = \sum_j |s_{ij}| \alpha_{ij} < \sum_j \alpha_{ij} = 1. \tag{18}$$

The entropy and corresponding effective number of active coordinates are descriptive diagnostics:

$$\mathcal{H}(\alpha_i) = -\sum_{j=1}^{d_g} \alpha_{ij} \log \alpha_{ij}, \qquad N_{\mathrm{eff}}(\alpha_i) = e^{\mathcal{H}(\alpha_i)}. \tag{19}$$

3.3 Carrier propagated through time

The recurrent carrier used by the next step and by the communication attention is

$$\bar{g}_i = \gamma_c \odot \mathrm{rms}(g_i), \qquad \gamma_c \in \mathbb{R}^{d_g} \text{ learnable and specific to the block.} \tag{20}$$

The coordinatewise gain γ_c is the learned gain of the carrier; the documented NOSCAL architecture does not insert additional scalar gates on the two large residual paths of the block. The causal recurrent computation at position i depends on positions $\leq i$ only.

4 Communication attention: reading the causal carrier

Collect the carrier vectors into a matrix:

$$\bar{G} = \begin{bmatrix} \bar{g}_1 \\ \vdots \\ \bar{g}_L \end{bmatrix} \in \mathbb{R}^{L \times d_g}. \tag{21}$$

Three learned projections generate multihead queries, keys and values:

$$Q^A = \bar{G} W_Q^A, \qquad K^A = \bar{G} W_K^A, \qquad V^A = \bar{G} W_V^A, \quad W_Q^A, W_K^A, W_V^A \in \mathbb{R}^{d_g \times d_g}. \tag{22}$$

Each of the $H = 8$ heads has $d_h = 64$ coordinates. For head a, use $q_i^a, k_i^a, v_i^a \in \mathbb{R}^{d_h}$ for its partition of Q^A, K^A, V^A.

4.1 Rotary position embedding (RoPE)

For positions i, RoPE applies a block diagonal matrix containing $d_h/2$ two-dimensional rotations, with frequencies

$$\omega_p = 10000^{-2p/d_h}, \qquad p = 0, \ldots, d_h/2 - 1, \tag{23}$$

$$R_i = \bigoplus_{p=0}^{d_h/2-1} \begin{bmatrix} \cos(i\omega_p) & -\sin(i\omega_p) \\ \sin(i\omega_p) & \cos(i\omega_p) \end{bmatrix}. \tag{24}$$

Equivalently, for row-vector convention, the rotated queries and keys are written $\tilde{q}_i^a = q_i^a R_i^\top$ and $\tilde{k}_i^a = k_i^a R_i^\top$. The rotation is applied to queries and keys, not to values.

4.2 Masked attention and communication residual

For every attention head a, the causal scores and simplex weights are

$$S_{ij}^{A,a} = \begin{cases} \dfrac{\tau_{A,a}}{\sqrt{d_h}} \langle \tilde{q}_i^a, \tilde{k}_j^a \rangle, & j \leq i, \\ -\infty, & j > i, \end{cases} \tag{25}$$

$$\rho_{ij}^{A,a} = \frac{\exp S_{ij}^{A,a}}{\sum_{\ell \leq i} \exp S_{i\ell}^{A,a}}, \qquad (\rho_{i1}^{A,a}, \ldots, \rho_{ii}^{A,a}) \in \Delta^{i-1}. \tag{26}$$

The notation $\rho^{A,a}$ is used for attention weights to avoid confusing them with the learned memory scalar β_{mem}. Here $\tau_{A,a}$ is the learned head-specific score factor.

Concatenate the eight weighted reads:

$$c_i^A = \underset{a=1}{\overset{H}{\mathrm{concat}}} \Big(\sum_{j \leq i} \rho_{ij}^{A,a} v_j^a \Big) \in \mathbb{R}^{d_g}. \tag{27}$$

Communication updates the hidden representation by

$$x_i^+ = x_i + c_i^A W_O, \qquad W_O \in \mathbb{R}^{d_g \times d_x}, \qquad (W_O)_{pq} \sim \mathcal{N}(0, 0.02^2). \tag{28}$$

There is no extra learned scalar multiplier on this residual. This is the documented NOSCAL communication path.

5 Stage-wise control: a second causal attention

The control mechanism C is *separate* from communication attention A: it has its own learned query, key and value projections, also with $H = 8$ heads of width $d_h = 64$, RoPE and causal masking. It is used to determine operator-mixture coefficients rather than the communicated residual.

Crucially, C is recomputed *at every geometric stage* on the currently transformed hidden sequence. If $x_i^{(k-1)}$ is the hidden state entering stage k, define

$$h_{k,i} = \mathrm{RMSNorm}_{d_x}\left(x_i^{(k-1)}\right), \qquad H_k = \begin{bmatrix} h_{k,1} \\ \vdots \\ h_{k,L} \end{bmatrix}. \tag{29}$$

Let the control attention have projection matrices $W_Q^C, W_K^C, W_V^C \in \mathbb{R}^{d_x \times d_g}$, and apply the same masked, RoPE-based attention construction as in Section 4 to H_k:

$$Q_k^C = H_k W_Q^C, \quad K_k^C = H_k W_K^C, \quad V_k^C = H_k W_V^C, \qquad c_{k,i}^C = C(H_k)_i \in \mathbb{R}^{d_g}. \tag{30}$$

The same *control-attention module* C is reused among the seven stages of a given block, although the attention weights are recomputed at each stage. Control modules of distinct blocks are independent. Thus the control at stage $k+1$ may read the effects of stages $1, \ldots, k$.

6 Lifting the current hidden state into operational space

The initial stage state is the communicated hidden:

$$x_i^{(0)} = x_i^+. \tag{31}$$

At each stage $k = 1, \ldots, K$, project its freshly normalized hidden state into the D-dimensional operational space:

$$u_{k,i} = \mathrm{SiLU}(h_{k,i} P_g) \odot (h_{k,i} P_\uparrow) \in \mathbb{R}^D, \tag{32}$$

where

$$P_g, P_\uparrow \in \mathbb{R}^{d_x \times D}, \qquad D = 4096. \tag{33}$$

Both projectors are *shared across all seven stages of the block*. P_g provides a nonlinear gate, while $P_\uparrow$ carries a linearly projected content vector. The Hadamard product forms the stage's operational representation. Importantly, the lifting is performed again for every stage after the previous update has returned to hidden space.

7 Continuous local routing on a six-slot simplex

Each stage has its own linear router head:

$$W_k \in \mathbb{R}^{d_g \times 6}, \qquad b_k \in \mathbb{R}^6. \tag{34}$$

From the stage-specific control read, it produces

$$\pi_{k,i} = \mathrm{softmax}(c^C_{k,i} W_k + b_k) \in \Delta^5. \tag{35}$$

The slots are indexed as follows: $j = 0$ is the *null identity* (zero deformation), while $j = 1, \ldots, 5$ correspond to five operators of the current family's type f_k. For $u \in \mathbb{R}^D$, define

$$\mathrm{op}_{k,0}(u) = 0, \qquad d_{k,i} = \sum_{j=1}^{5} \pi_{k,i,j}\ \mathrm{op}^{(f_k)}_{k,j}(u_{k,i}) \in \mathbb{R}^D. \tag{36}$$

The mixture is continuous; there is no Gumbel hard selection in this version. In the idealized boundary case $\pi_{k,i,0} = 1$, all other coefficients vanish and hence $d_{k,i} = 0$ exactly. A finite softmax logit vector typically produces strictly positive probabilities, so this identity statement is a property of the simplex boundary, not a claim that exact one-hot output is routinely reached during training.

No single global catalogue of all operators is used. Each of the seven stage occurrences owns the parameters of its five local operators. In particular, the three M stages in (1) are independent.

8 Operator families in the lifted space

Each operator maps $\mathbb{R}^D \to \mathbb{R}^D$ using rank-r learned frames, with $D = 4096$ and $r = 32$. Unless otherwise indicated, operations on intermediate r-vectors are elementwise. The subscripts k, j on frame matrices are temporarily suppressed for readability, but all frames belong to their *specific stage and operator slot.*

8.1 F: free low-rank nonlinear field

For the five operators of a stage of family F, let

$$A^F_j, B^F_j, U^F_j \in \mathbb{R}^{D\times r}. \tag{37}$$

Then

$$\boxed{F_j(u) = [\mathrm{SiLU}(uA^F_j) \odot (uB^F_j)](U^F_j)^\top.} \tag{38}$$

No fixed internal catalogue of elementary functions is imposed on the five free fields: their diversity comes from learned frames. As $t \to 0$, $\mathrm{SiLU}(t) = \frac{1}{2}t + O(t^2)$, so the leading local term is quadratic:

$$F_j(u) = \tfrac{1}{2}[(uA^F_j) \odot (uB^F_j)](U^F_j)^\top + O(\|u\|^3). \tag{39}$$

The documented initialization multiplies the initial B^F_j frame by `F_INIT_GAIN` $= 6.0$ for empirical matching of the deformation magnitude to the other families; this scaling is an initialization rule, not an additional runtime operation.

8.2 B: tangent-to-orthogonal bending

For each bending slot, take

$$T_j, N_j \in \mathbb{R}^{D\times r}. \tag{40}$$

Let Q_{T_j} be an orthonormal basis obtained from the QR factorization of T_j. The second frame is orthogonalized against the span of T_j:

$$N^\perp_j = N_j - Q_{T_j}(Q^\top_{T_j} N_j). \tag{41}$$

The resulting operator is

$$\boxed{B_j(u) = \mathrm{SiLU}(uT_j)(N^\perp_j)^\top.} \tag{42}$$

$N_j^{\perp}$ is not normalized again. If the frame is rank-deficient, the intended projection is onto the actual column span of T_j; the QR expression above is the documented construction for the frame-based implementation.

8.3 M: point-dependent metric-style deformation

For each metric-family operator, use

$$U_j^M, A_j^M \in \mathbb{R}^{D\times r}, \qquad \boxed{M_j(u) = [(uU_j^M) \odot \phi_j(uA_j^M)](U_j^M)^\top.} \tag{43}$$

The five local nonlinearities are

$$\begin{aligned} &\phi_1(q) = \tanh(q), && \phi_2(q) = \mathrm{softsign}(q) = \frac{q}{1+|q|}, \\ &\phi_3(q) = \mathrm{asinh}(q), && \phi_4(q) = q \odot \sigma(\beta_{M,4}|q|), \\ &\phi_5(q) = \mathrm{SiLU}(q). \end{aligned} \tag{44}$$

The subscript on $\beta_{M,4}$ distinguishes this metric-family parameter from the memory-orientation scalar β_{mem}. The M family is named for its learned, point-dependent modifications, but the formulas alone do *not* assert that a positive-definite Riemannian metric tensor is explicitly constructed.

8.4 V: oscillatory deformation family

This family has the same low-rank frame structure as M:

$$U_j^V, A_j^V \in \mathbb{R}^{D\times r}, \qquad \boxed{V_j(u) = [(uU_j^V) \odot \phi_j(uA_j^V)](U_j^V)^\top.} \tag{45}$$

The five scalar/vector-valued laws are

$$\begin{aligned} &\phi_1(q) = \tanh(q), && \phi_2(q) = \mathrm{asinh}(q), \\ &\phi_3(q) = \sin(\omega \odot q), && \phi_4(q) = \mathbf{1} - \cos(\nu \odot q), \\ &\phi_5(q) = c_s \odot \sin(\omega \odot q) + c_c \odot [\mathbf{1} - \cos(\nu \odot q)]. \end{aligned} \tag{46}$$

The necessary frequency and combination vectors $\omega, \nu, c_s, c_c \in \mathbb{R}^r$ are learned wherever the associated law uses them. Every oscillatory operator slot has its own applicable parameters; the notation above suppresses the slot index.

8.5 S: linear shear family

For $U_j^S, V_j^S \in \mathbb{R}^{D\times r}$, define

$$\boxed{S_j(u) = (uV_j^S)(U_j^S)^\top.} \tag{47}$$

This is a general learned low-rank linear map of the operational vector. The term *shear* is the name of this family in CASC; the learned map is not restricted by the equations to a classical volume-preserving shear matrix.

8.6 Family-dependent return nonlinearity and initialization

Before being translated back to hidden space, the mixed deformation uses

$$\psi_F = \psi_B = \psi_M = \psi_V = \mathrm{SiLU}, \qquad \psi_S = \mathrm{Id}\,. \tag{48}$$

The documented frame initialization starts from normal frame entries with variance approximately $1/D$. Certain value-contributing frames are initialized with an additional factor $\varepsilon_0 = 0.02$, so typical initial relative deformation magnitudes are in the 10^{-4}–10^{-3} range. This is an implementation initialization prescription, not a constraint on the trained operator norms.

9 Geometric cascade: composition through hidden space

The crucial structural feature of CASC–NTP is that a deformation is *immediately returned to the hidden state* before the next deformation is evaluated. The shared return projector is

$$P_{\downarrow} \in \mathbb{R}^{D \times d_x}, \tag{49}$$

independent of $P_{\uparrow}$; the architecture does not require $P_{\downarrow} = P_{\uparrow}^{\top}$.

For every stage k and token position i, the complete state transition is

$$x_i^{(0)} = x_i^{+}, \tag{50}$$
$$h_{k,i} = \mathrm{RMSNorm}_{d_x}(x_i^{(k-1)}), \tag{51}$$
$$u_{k,i} = \mathrm{SiLU}(h_{k,i}P_g) \odot (h_{k,i}P_{\uparrow}), \tag{52}$$
$$c_{k,i}^{C} = C(H_k)_i, \tag{53}$$
$$\pi_{k,i} = \mathrm{softmax}(c_{k,i}^{C}W_k + b_k), \tag{54}$$
$$d_{k,i} = \sum_{j=1}^{5} \pi_{k,i,j}\ \mathrm{op}_{k,j}^{(f_k)}(u_{k,i}), \tag{55}$$
$$\Delta x_{k,i} = \psi_{f_k}(d_{k,i})P_{\downarrow}, \tag{56}$$
$$\boxed{x_i^{(k)} = x_i^{(k-1)} + \Delta x_{k,i}.} \tag{57}$$

In these equations, H_k contains the normalized current states of *all* sequence positions, and C is causally masked. The output of the block is $B_b(x)_i = x_i^{(K)}$. This defines the deterministic forward computation of the entire block, conditional on its parameters.

The operative cycle is

$$\underbrace{x^{(k-1)}}_{\text{hidden}} \longrightarrow \underbrace{h_k}_{\text{renormalize}} \longrightarrow \underbrace{u_k}_{\text{lift}} \longrightarrow \underbrace{d_k}_{\text{routed geometry}} \longrightarrow \underbrace{x^{(k)}}_{\text{return and add}}. \tag{58}$$

The control computation is evaluated from h_k at the same stage and modulates d_k. This is *not* a single sum of seven operators applied once to the initial u: both u_k and c_k^C change when earlier stage updates change the hidden state.

10 Full forward map and autoregressive readout

Combining the previous sections, the causal language model is

$$(y_1, \ldots, y_L) \xrightarrow{E} X^{[0]} \xrightarrow{B_1} \cdots \xrightarrow{B_S} X^{[S]} \xrightarrow{\mathrm{RMSNorm}_{d_x}} H^{\mathrm{out}} \xrightarrow{E^{\top}} \Lambda \in \mathbb{R}^{L \times V}. \tag{59}$$

For each position,

$$h_i^{\mathrm{out}} = \mathrm{RMSNorm}_{d_x}(x_i^{[S]}), \tag{60}$$
$$\ell_i = h_i^{\mathrm{out}}E^{\top} \in \mathbb{R}^{V}, \tag{61}$$
$$p_i(v) = \frac{\exp \ell_{i,v}}{\sum_{w=0}^{V-1} \exp \ell_{i,w}}, \qquad v = 0, \ldots, V-1. \tag{62}$$

The tied projection uses a *single* trainable E for the token embedding and the readout, rather than maintaining independent embedding and output-head matrices.

10.1 Causality

Each $x_i^{[S]}$ depends only on tokens at positions at most i: the GRU state is causal, both attentions use triangular masks, and the geometric operations are pointwise in position once the causal control vectors are constructed. Therefore the logits ℓ_i define a next-token conditional distribution $p_\theta(y_{i+1} \mid y_{\le i})$.

11 The three probability simplices

The architecture contains three conceptually different normalization mechanisms:

Distribution	Space	Role
α_i	$\Delta^{d_g-1} = \Delta^{511}$	Coordinate-weight distribution used to structure the signed recurrent memory.
$\rho_{i\cdot}^{A,a}$ and $\rho_{k,i\cdot}^{C,a}$	Δ^{i-1}	Distributions over accessible past and current sequence positions for each head.
$\pi_{k,i}$	Δ^5	Distribution over the null-identity slot and five operators of the current family.

The meanings are not interchangeable: α distributes carrier coordinates, attention weights distribute source positions, and π distributes stage-level deformation choices.

12 Learning objective and actual optimization mechanism

This section distinguishes the trainable machine from the optimizer and states *the training mechanism documented for CASC–NTP*. The model is trained end-to-end using ordinary next-token cross-entropy. For position mask $M_i \in \{0, 1\}$,

$$\boxed{\mathcal{L}_{\mathrm{NTP}}(\theta) = -\frac{1}{\sum_i M_i} \sum_i M_i \log p_{\theta,i}(y_{i+1}).} \tag{63}$$

Here θ comprises the embedding, recurrent memory, both attention mechanisms, local routing heads, operator frames, projectors, and normalization parameters.

12.1 Standard backward propagation

Gradients are computed by **classical backpropagation**, i.e. reverse-mode differentiation through the full forward computation graph (including attention, recurrent states, stage-wise geometric updates, and tied lexical weights). With $z_{i,v} = \ell_{i,v}$ and one-hot target $e_{y_{i+1}}$,

$$\frac{\partial \mathcal{L}_{\mathrm{NTP}}}{\partial z_{i,v}} = \frac{M_i}{\sum_j M_j} \Big[p_{\theta,i}(v) - \mathbf{1}\{v = y_{i+1}\} \Big]. \tag{64}$$

For any trainable parameter θ_a, the actual update gradient is

$$g_{t,a} = \frac{\partial \mathcal{L}_{\mathrm{NTP}}(\theta_t)}{\partial \theta_a}, \tag{65}$$

obtained via the chain rule throughout the network. In particular, the coefficients $\pi_{k,i,j}$ route forward information and receive gradients through (35)–(55); they *do not replace* reverse-mode derivatives as an optimization rule.

12.2 AdamW parameter updates

The documented trainer uses AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, decoupled weight decay $\lambda = 0.01$, and global gradient-norm clipping at 1.0. For gradient g_t after any prescribed clipping, the standard AdamW equations are

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \tag{66}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)(g_t \odot g_t), \tag{67}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \tag{68}$$

$$\theta_t = (1-\eta_t\lambda)\theta_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon_{\mathrm{AdamW}}}. \tag{69}$$

These are conventional optimizer equations supplied here to identify the training mechanism; $\varepsilon_{\mathrm{AdamW}}$ is the optimizer stabilization constant and is *not* the $\varepsilon = 10^{-6}$ used in the geometric normalizations. The released trainer uses the PyTorch default $\varepsilon_{\mathrm{AdamW}} = 10^{-8}$. The learning-rate schedule belongs to the training protocol, not to the mathematical definition of the machine.

The published run, NTP_A_4B, used an initial learning rate of 6×10^{-4} after a warmup of 2000 steps and a warmup–stable–decay schedule with a linear decay over the last 15% of training. It completed 49,215 optimizer steps over 3.592×10^9 prediction tokens, one full pass over its corpus. These numbers document that run; they are not requirements for constructing or evaluating CASC–NTP.

13 Parameter count and sharing rules

The full-size configuration has 175,218,812 trainable parameters, partitioned as follows:

Component	Trainable parameters
Tied embedding E	65.536 M
Causal memory and attention mechanisms	19.931 M
Shared within-block operational projectors	50.332 M
Geometric operator-family frames	39.325 M
Stage routers and normalization scales	0.095 M
Total	**175.219 M**

The three large projection matrices $P_g, P_\uparrow, P_\downarrow$ contribute, per block,

$$3 d_x D = 3(1024)(4096) = 12{,}582{,}912 \tag{70}$$

weights. Across four blocks, this equals 50,331,648. The tied embedding has $V d_x = 65{,}536{,}000$ parameters. Component totals are rounded to three decimal places in millions.

Parameter sharing is precisely localized: the attention C and operational projectors are reused across stages of a *single* block; operator-frame parameters and router head parameters are distinct for each stage; blocks are separate; and lexical weights are tied between input embedding and final readout.

14 Architectural diagnostics and interpretation

The trainer logs the following diagnostics per block and geometric stage, without treating them as extra losses:

$$\pi_{k,i,0}, \qquad \exp \mathcal{H}(\pi_{k,i}), \qquad \frac{\|d_{k,i}\|_2}{\|u_{k,i}\|_2}, \qquad \frac{\|\Delta x_{k,i}\|_2}{\|x_i^{(k-1)}\|_2}. \tag{71}$$

For the cumulative dynamics, relevant quantities include

$$\cos(\Delta x_{k,i}, \Delta x_{\ell,i}), \qquad \frac{\|\sum_{j\le k} \Delta x_{j,i}\|_2}{\|x_i^+\|_2}, \qquad \frac{\|u_{k,i} - u_{k-1,i}\|_2}{\|u_{k-1,i}\|_2}, \qquad \cos(u_{k,i}, u_{k-1,i}), \tag{72}$$

where cosine similarity is defined when its arguments have nonzero norm, and denominators should be numerically safeguarded when these diagnostics are implemented. Percentile-based statistics of deformation magnitudes (e.g. r_{90} and r_{99} in the training log) and norms of the F family's intermediate factors are also useful to identify concentration, cancellation, stage inactivity, or uncontrolled initialization growth.

14.1 Meaning and limits of the geometric vocabulary

The terms *geometric deformation*, *metric*, *bending*, *shear*, and *manifold* describe the architecture's intended interpretation: learned operators move a hidden representation through a sequence of locally defined transformations, and each update changes the inputs to the subsequent stage. The equations establish this *computational* interpretation. They do not, by themselves, prove that the hidden-state data occupy a smooth embedded submanifold, that any learned update is a diffeomorphism, that all mappings are invertible, or that the M family defines a positive-definite metric. Those would require additional mathematical assumptions and separate verification. In particular, the low-rank, gated, residual maps are not constrained here to preserve dimension, distance, topology, volume, or curvature in a specified formal sense.

15 Compact specification of the complete machine

For $b = 1, \dots, S$, the block acts as

$$\boxed{\begin{aligned}
&\text{recurrent signed-simplex carrier } \bar{G}^{(b)} \xrightarrow{A^{(b)}} c^{A,(b)} \longrightarrow x^{+,(b)},\\
&\qquad x^{(0,b)} = x^{+,(b)},\\
&\left.\begin{aligned}
h_k^{(b)} &= \mathrm{RMSNorm}\big(x^{(k-1,b)}\big),\\
u_k^{(b)} &= \mathrm{SiLU}\big(h_k^{(b)} P_g^{(b)}\big) \odot \big(h_k^{(b)} P_\uparrow^{(b)}\big),\\
c_k^{C,(b)} &= C^{(b)}\big(h_k^{(b)}\big),\\
\pi_k^{(b)} &= \mathrm{softmax}\big(c_k^{C,(b)} W_k^{(b)} + b_k^{(b)}\big),\\
d_k^{(b)} &= \textstyle\sum_{j=1}^{5} \pi_{k,j}^{(b)}\, \mathrm{op}_{k,j}^{(f_k,b)}\big(u_k^{(b)}\big),\\
x^{(k,b)} &= x^{(k-1,b)} + \psi_{f_k}\big(d_k^{(b)}\big) P_\downarrow^{(b)}
\end{aligned}\right\} \quad k = 1, \dots, 7.
\end{aligned}} \tag{73}$$

All operations in this display are performed over token positions as described earlier, with causal dependencies preserved by the recurrent update and masked attentions. Finally,

$$\boxed{\begin{aligned}
X^{[0]} &= E[y_1, \dots, y_L],\\
X^{[S]} &= B_S \circ \cdots \circ B_1\big(X^{[0]}\big),\\
\Lambda &= \mathrm{RMSNorm}_{d_x}\big(X^{[S]}\big) E^\top,\\
p_\theta(y_{i+1} \mid y_{\le i}) &= \mathrm{softmax}(\Lambda_i)_{y_{i+1}},\\
g_\theta &= \nabla_\theta \mathcal{L}_{\mathrm{NTP}} \quad \text{(classical backpropagation)},\\
\theta_{t+1} &= \mathrm{AdamW}(\theta_t, g_\theta).
\end{aligned}} \tag{74}$$

Verification against the released code

The statements of this document were checked against the released source code (`casc.py`, `train.py`) with `test_casc.py`:

- tensor shapes and the parameter count: 175,218,812 trainable parameters in 274 state-dict entries, whose names and shapes are exactly those of the published checkpoint (loaded with `strict=True`);
- parameter sharing: one control attention C and one projector triplet $(P_g, P_\uparrow, P_\downarrow)$ per block, independent operator frames and router heads for every stage, a single tied matrix E;
- the equations of Sections 3–10, by independent re-implementation: signed-simplex GRU, causal attentions, identity slot, the F, B, M, V, S operators, ψ per family, renormalization and control recomputed at every stage, causality of the whole machine;
- the optimizer: `torch.optim.AdamW` with $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\lambda = 0.01$ and the PyTorch default $\varepsilon_{\mathrm{AdamW}} = 10^{-8}$, global gradient-norm clipping at 1.0.

The measurements of the published run are reported in `report.html`.